

Dataset Overview

File	large_test_dataset.csv
Total Rows	100
Total Columns	12
Numeric Columns	4
Text Columns	7
Missing Data	0%
Duplicates Removed	0

Numeric Summary

Column	Mean	Median	Std	Min	Max	Outliers
id	50.5	50.5	29.01	1.0	100.0	0
rating	4.35	5.0	1.07	1.0	5.0	9
price	148.59	74.99	264.24	8.99	1799.99	12
helpful_votes	447.14	345.0	330.63	78.0	2345.0	5

Sentiment Analysis

Text Column	review
Total Samples	100
Positive	92.0%
Negative	8.0%
Neutral	0.0%

ML Model Results

Best Model	Complement NB
Accuracy	95.0%
F1 Score	94.2%
CV Mean	89.3%
CV Std	+/- 1.41%
Train Size	80
Test Size	20
Total Models Trained	4

Model Comparison Table

Model	Acc%	Prec%	Rec%	F1%	CV%	Time(s)
Logistic Regression	85.0	72.2	85.0	78.1	79.8	0.65

SGD Classifier	90.0	92.5	90.0	90.0	85.3	1.25
Complement NB	95.0	95.3	95.0	94.2	89.3	0.48
Linear SVM	85.0	72.2	85.0	78.1	83.5	0.25

AI-Generated Insights

* Dataset Overview

Your dataset has 100 rows and 12 columns. Lumiq cleaned it automatically.

* rating is Skewed

The 'rating' column is skewed to the left (skewness=-1.81). Consider log transformation.

* Outliers in rating

9 outliers detected in 'rating' using the IQR method.

* Mostly Positive Sentiment

92.0% of 100 entries in 'review' are positive. Excellent reception!

* Best Model: Complement NB

Achieved 95.0% accuracy and 94.2% F1 score with 89.3% cross-validation mean.

* Top Keywords Found

Most important terms: "quality, "good, "excellent. These drive the overall sentiment.